



Fazeel Nizam

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Kalutara, Sri Lanka

<https://fazeelnizam.github.io/portfolio/>

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Profile

Computer Engineer with hands-on experience in Embedded Systems, Edge AI, FPGA, and IoT. Proven ability to architect end-to-end systems, from firmware and hardware integration to cloud connectivity and computer vision pipelines. Seeking opportunities to design cutting-edge Embedded, IoT, AI/ML, and FPGA systems that bridge research and real-world applications.

Technical Skills

Embedded & Hardware,

NVIDIA Jetson, Raspberry Pi, FPGA (Xilinx Artix-7/Spartan6), ESP32, Arduino, PCB Design

AI/ML, Computer Vision, OpenCV, OCR, Edge Computing, CNN, YOLO

Protocols, MQTT, LoRa, SDI-12, SPI, I²C, UART, BLE, HTTP/REST, JSON, RTSP, WebSockets

Programming & Scripting, Python, VHDL, Java, Embedded C/C++, CUDA, JavaScript (ES6)

Software & Tools,

Vivado, EasyEDA, GStreamer, Proteus, MATLAB, Git, AutoCAD, Figma, AWS IoT Core, CI/CD

Web Technologies, ReactJS, NextJS, HTML5, CSS3, Tailwind, MUI, FastAPI, PostgreSQL, Nginx

Professional Experience

Freelance Embedded System Engineer,

SLTMobitel Digital Projects

May 2026 – Present

Colombo, Sri Lanka

- Leading the end-to-end development of **SecureGate**, an enterprise IoT access control system, encompassing custom PCB hardware, device firmware, a FastAPI backend, a React frontend, cloud hosting, and CI/CD pipelines.
- Architected a multi-VM infrastructure utilizing an Nginx-powered load balancer, a dedicated MQTT broker, and a unified Application VM hosting both the frontend and backend.
- Developed secure end-to-end communication between the FastAPI backend, React frontend, and ESP32 nodes by implementing RSA and dynamic key exchange protocols.

Trainee IoT Engineer, SLTMobitel Digital Projects

Oct 2025 – Apr 2026

Colombo, Sri Lanka

- Developing an AI-powered parking management system by implementing low-latency RTSP video pipelines using GStreamer on WSL2 to bridge legacy CCTV infrastructure with modern AI processing.

- Designing a scalable IoT ecosystem featuring smart access control, electrical power monitoring, and automated HVAC control driven by real-time CO2, temperature, and humidity data analysis.

Technical Assistant, *Career Guidance Unit, OUSL*

Mar 2025 – Sep 2025

Nawala, Sri Lanka

- Engineered a multi-node environmental monitoring system and a portable Greenhouse Gas (GHG) Analyzer for agricultural research.
- Provide technical support for university-wide hybrid career events.

Trainee Electronic Engineer,

Jul 2024 – Dec 2024

Arthur C Clarke Institute for Modern Technologies

Katubedda, Sri Lanka

- Designed an ESP32-based Nano-Sterlite Orbital Magnetic Field Simulation Device (Helmholtz Cage), utilizing a PID algorithm to stabilize a frictionless air-bearing platform.
- Implemented FreeRTOS for precise task scheduling to manage magnetic field generation and sensor feedback loops.
- Developed a wireless data logging system with an embedded web dashboard (hosted via SPIFFS) for remote parameter control.

Technical Assistant, *Neo Space Lab, OUSL*

Sep 2023 – Dec 2023

Nawala, Sri Lanka

- Designed UI/UX for a Research Publication Web App and IoT dashboards using NextJS and MUI.
- GitHub Repositories: *IoT Weight Measurement System* [↗](#), *LKO Web Dashboard* [↗](#)

Banking Trainee,

Aug 2022 – Dec 2022

Commercial Bank of Ceylon, Main Branch Wellawatta

- Handled lower-counter operations, including account opening, debit card issuance, and E-Remittances, ensuring compliance and customer satisfaction

Education

BSc (Hons) in Computer Engineering,

2021 – 2026

The Open University of Sri Lanka

Current GPA: 3.58

Focus: Embedded Systems, AI/ML, IoT, Digital System Design, Computer Vision, Processor Design

Featured Projects

Automatic Number Plate Recognition (ANPR) and Vehicle Park Access Control System, *Tech Stack:*

2026

Python, ReactJS, Computer Vision, OCR, YOLO, MQTT

- Developing an end-to-end ANPR system, focusing heavily on OCR optimization and system architecture to ensure accurate, real-time plate reading.
- Actively training and fine-tuning models to recognize highly variable localized data, specifically targeting the full spectrum of Sri Lankan number plates, including older, non-standard, and new custom formats.

Fuzzy Matrix Processing Unit (University Project)

2025

Tech Stack: VHDL | Xilinx Vivado | FPGA (Artix-7)

- Architected a dedicated hardware accelerator using VHDL to perform Fuzzy Matrix operations.
- Validated design integrity through testbenches and post-synthesis timing simulations in Xilinx Vivado.
- Implemented and verified the synthesized design on a Nexys A7 Development Board (Artix-7 FPGA), demonstrating real-world hardware functionality.

Multi-Unit Environmental & Agricultural Monitoring

2025

System (On Going) , *Tech Stack: ESP32 | Pi Zero |*

Embedded C | Python | MQTT | SDI-12

- Architected a wireless sensor network with three coordinated ESP32 nodes for real-time soil (TEROS 10/22), environmental, and weather (ATMOS 41) monitoring.
- Designed a robust data pipeline where edge nodes serve sensor data via ESPNow, consumed by a central aggregator for upload via a MQTT Broker.
- Programmed a self-sufficient weather station capable of parsing complex SDI-12 protocols and logging timestamped data.
- Developed the system's backend architecture using Python FastAPI and created the frontend dashboard using Next.js.

Portable GHG Analyzer for Field Research

2025

Tech Stack: ESP32 | Embedded C | SPIFFS

- Built a standalone, keypad-operated device for precision Greenhouse Gas measurement, significantly improving the efficiency of agricultural field research.
- Developed automated control algorithms for air circulation and stabilization, ensuring high-fidelity readings in varying outdoor conditions.

✦ Extra Curricular Activities

Student Volunteer , *USBUS Mobile American Space Program 2025*

President , *ENOSPIRE OUSL 2025 Entrepreneurship Day Organizing Committee*

Organizing Committee Member , *TalentLyft '25 Career Fair, OUSL*

Technical Team Lead , *TalentLyft '23 Career Fair, OUSL*

Volunteer Teacher & Committee Member, *SVP Educational Institute, Kalutara*

🔗 References

Prof. Channa J Kumarage, *Computer Information Systems Professor,*
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